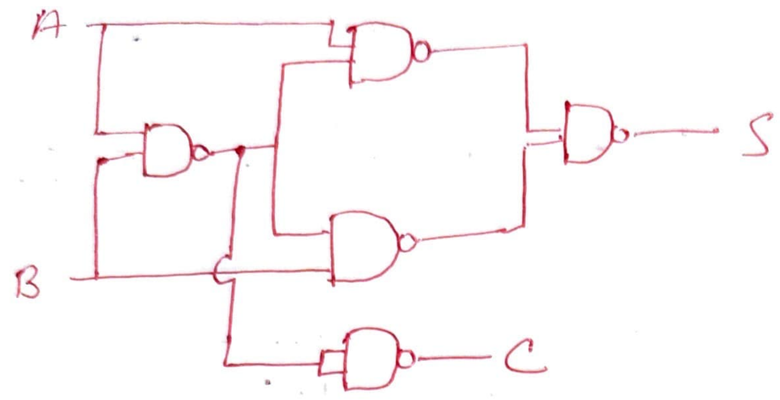
A - 3 Solution

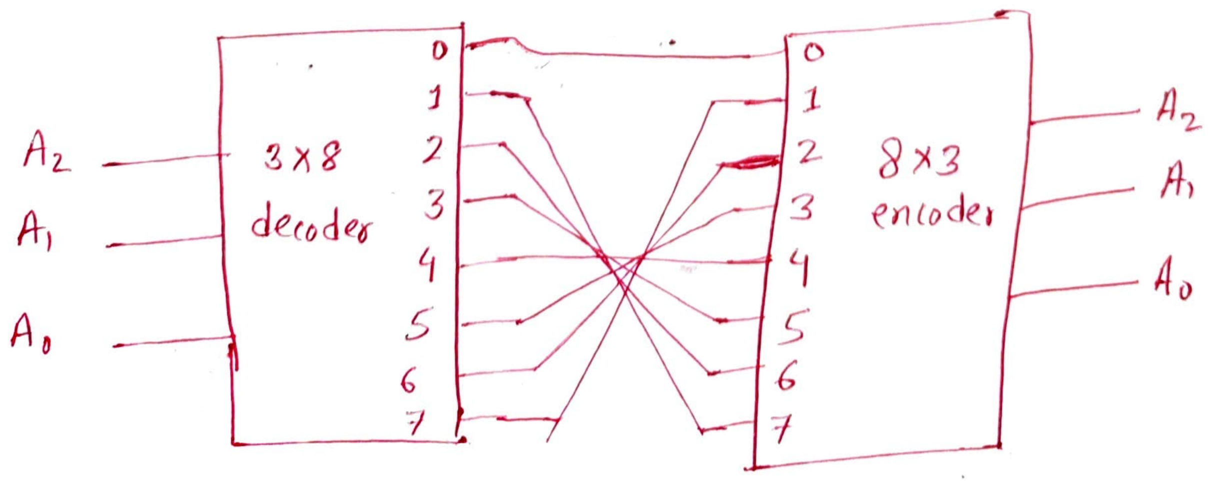
1) Half Adder using NAND



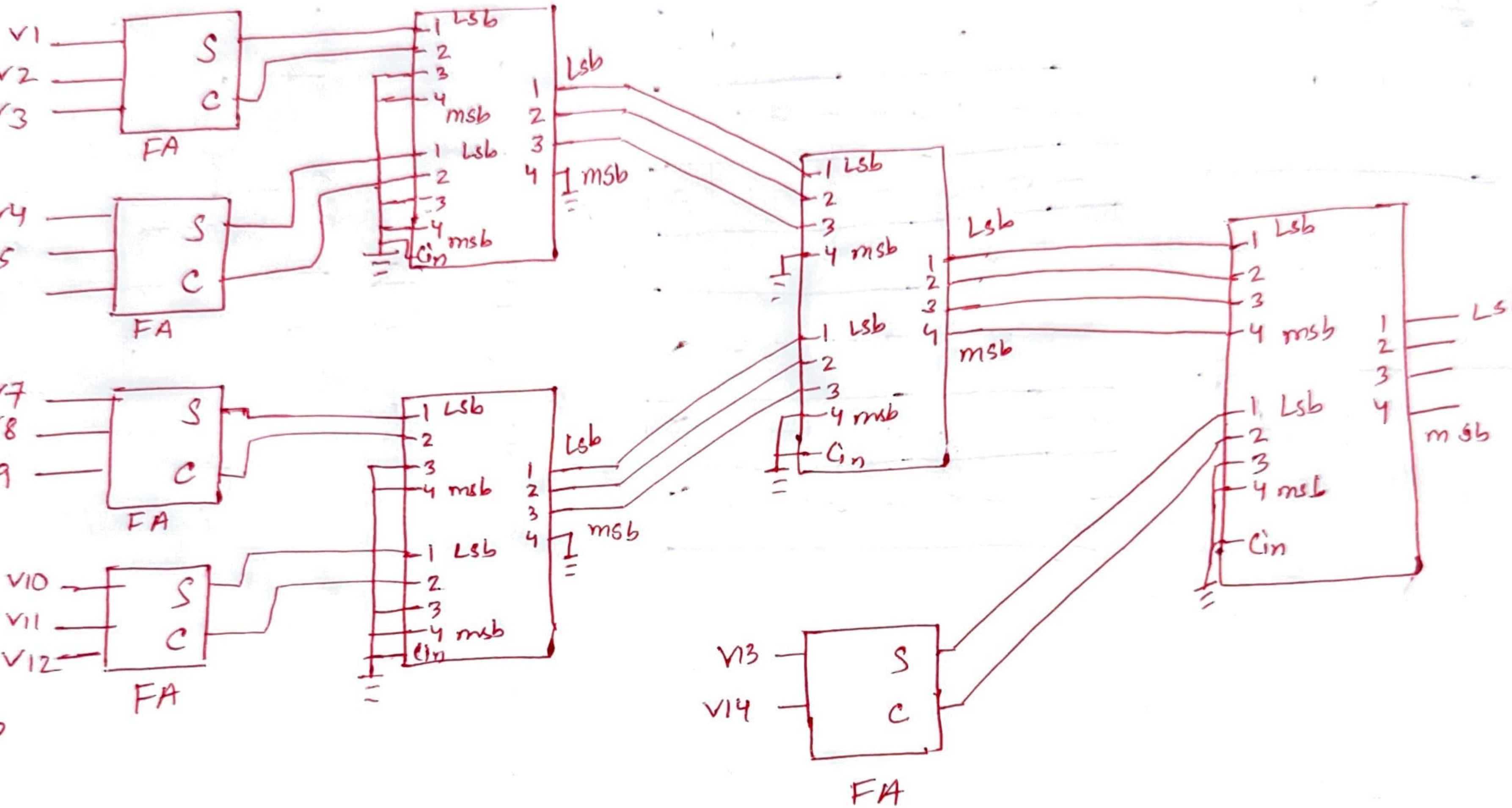
2) 2's ampliment to binary

Truth Table

A_2	A_1	A_0	A_2	A_1	A_0
0	0	0	0	0	0
0	0	1	1	1	1
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	0	0
1	0	1	0	1	1
1	1	0	0	1	0
1	1	1	0	0	1

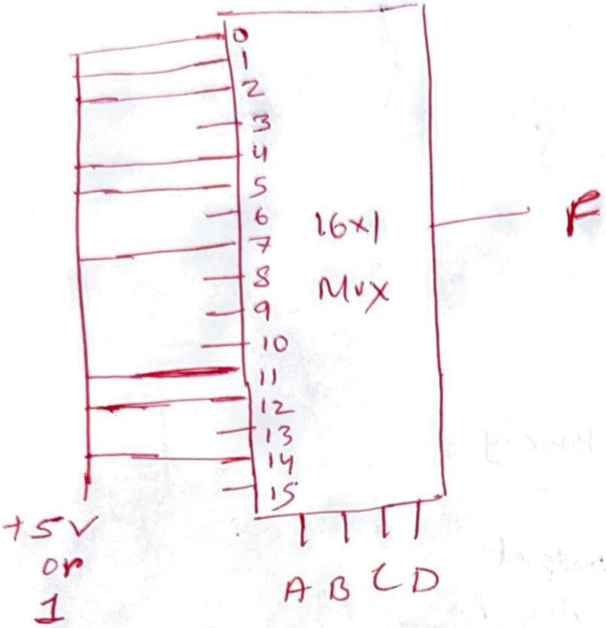


3) 14-person voting system



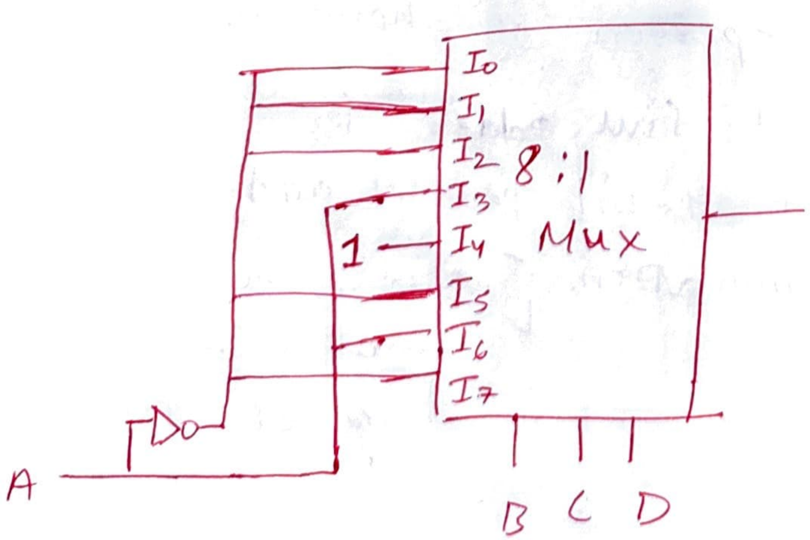
4) $F(A,B,C,D) = \Sigma(0,1,2,4,5,7,11,12,14)$

a) 16:1 Mux



b) Single 8:1 Mux

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
	A'	A'	A'	A	1	A'	A	A'



Must use external gates for A'.

1c) single 4:1 Mux.

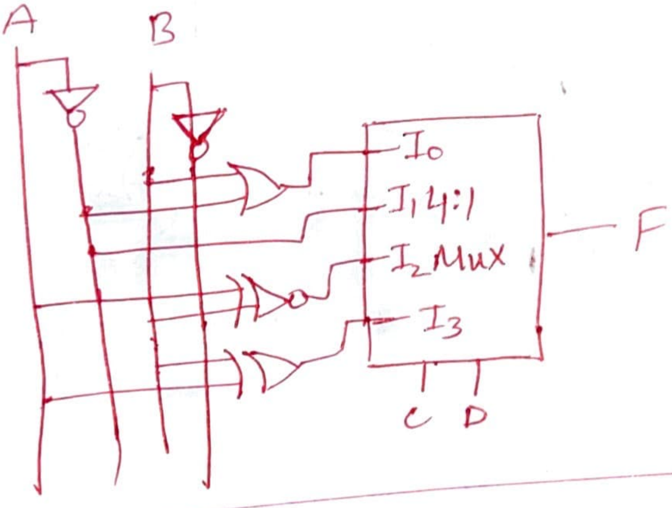
	I_0	I_1	I_2	I_3
$A'B'$	0	1	2	3
$A'B$	4	5	6	7
AB'	8	9	10	11
AB	12	13	14	15

$$I_0 = A'B' + A'B + AB$$
$$= A'(B'+B) + AB$$
$$= A' + AB$$
$$= (A'+A)(A'+B)$$
$$= A'+B$$

$$I_1 = A'B' + A'B$$
$$= A'(B'+B)$$
$$= A'$$

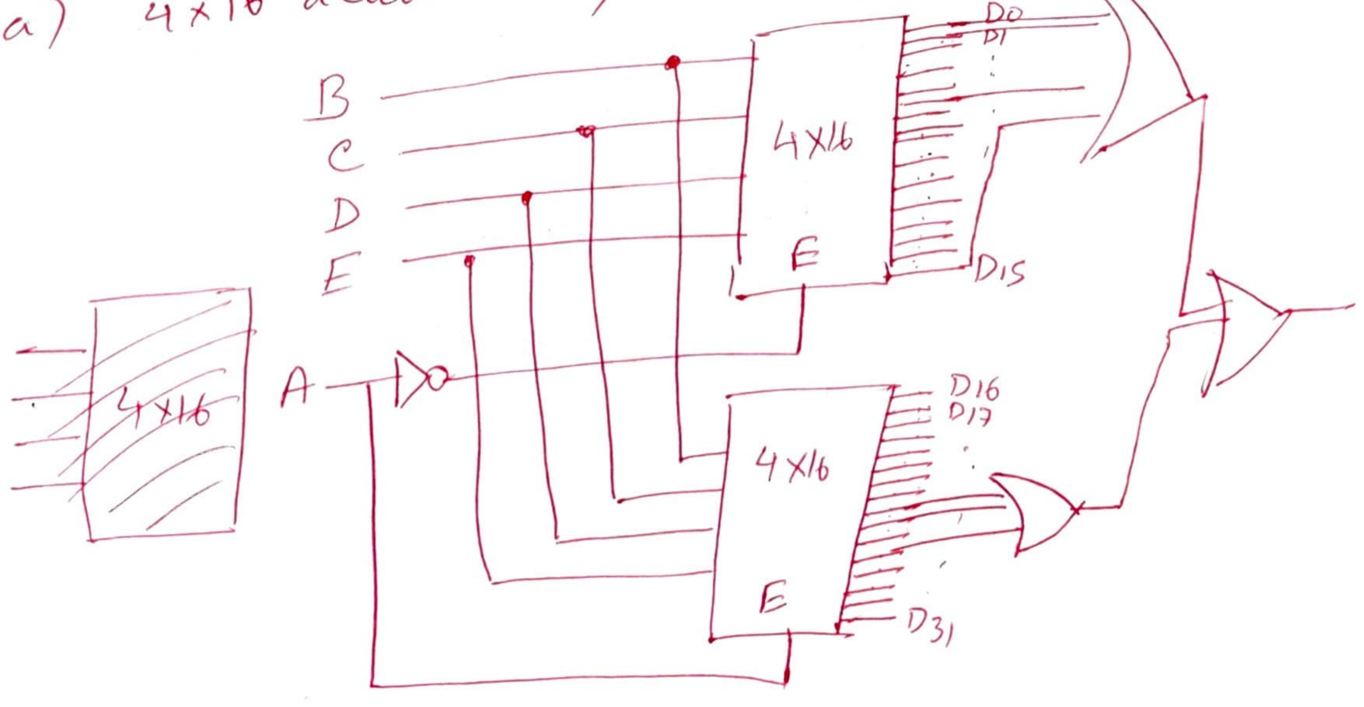
$$I_2 = A'B' + AB$$
$$= A \odot B$$

$$I_3 = A'B + AB'$$
$$= A \oplus B$$



5) $F(A, B, C, D, E) = \Sigma(0, 1, 5, 15, 24, 25, 27)$

a) 4×16 decoder only



5b) 2x4 decoders only